SIMATSENGINEERING

ASSESSMENTTOOL2:ResearchReview&Analysis

COURSE NAME: **ARTIFICIALINTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED

CO4: Developintelligentsoftwareagentsusingsuitableagentarchitecturesand learning techniques.(BTL 3)

AssessmentToolWeightage:30%

TotalMarks:100 **Time:90Mins**

No. of Questions: 1

AnnexureA

ResearchReview&Analysis

Code of Conduct:

I Kamalli shree V(192525030)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11.ResearchPaperQualityandReferencing	10%	
12.TechnicalPresentationandResearchDefense	5%	
Total	100%	

SignatureoftheFaculty TotalMarksAwarded

ANSWERANYONEQUESTION

S.No.	ResearchReview&AnalysisQuestion	PrimaryTopic
1	**DecisionTreeLearning:** Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance.Identify a research gap and suggest a possible improvement.	Decision Trees
2	**Neural Networks vs Statistical Learning:** Reviewandcompareresearchonneural-networkand statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	**Kernel Methods:** Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluatetheiradvantagesandlimitationscomparedwith neural-network-based approaches.	Kernel Methods
4	**ReinforcementLearning:** Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	**Explainable and Generalizable AI:** Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify currentlimitations and propose future research directions.	Inductive &Explanation-Base d Learning

Structureofthe Report:

S.No.	Paper Section	Expected Content
1	**Title**	Specific and concise title related to the selected research question/topic.
2	**Abstract**	Briefsummaryoftheresearchproblem,reviewmethodology,major findings, research gap, and conclusion.
3	**Keywords**	4–6 important keywords related to the selected topic.
4	**1. Introduction**	IntroducetheAItopic,background,importance,motivation,andscopeof the review.
5	**2. Research Problemand Objectives**	Clearlystate the research problem and specific objectives of the review.
6	**3.Literature Review**	Reviewrelevantresearchpapersandsummarizemajorapproaches, methodologies, findings, and contributions.
7	**4.Comparative Analysis**	Compare the reviewed approaches based on suitable parameters such as accuracy,complexity,interpretability,computationalcost,scalability,and generalization.
8	**5.Critical Analysis**	Discussstrengths,weaknesses,limitations,assumptions,andpractical challenges identified across the reviewed studies.
9	**6.Research Gap**	Identifyunresolvedproblems,limitationsinexistingresearch,andareas requiring further investigation.
10	**7.Proposed Research Direction**	Suggestpossiblesolutions,improvements,orfutureresearchdirections based on the identified gap.
11	**8. Discussion**	Discusstheoverallinsightsobtainedfromthereviewandtheirimplications for AI research and applications.
12	**9. Conclusion**	Summarizethemajorfindings,keyinsights,researchgap,andfuture direction.
13	**References**	Listallresearchpapers,books,datasets,websites,andothersourcesusinga consistent citation style.

Neural Networks vs Statistical Learning: A Comparative Review of Prediction, Classification, Interpretability, and Computational Performance

Abstract

Neural networks and statistical learning methods are two important approaches used for prediction and classification in artificial intelligence and data analysis. While statistical learning methods provide established mathematical models with strong interpretability and relatively lower computational requirements, neural networks can model complex nonlinear relationships and often benefit from large and high-dimensional datasets. This research paper reviews and compares these two approaches based on prediction performance, classification capability, data requirements, generalization, interpretability, computational cost, and suitability for different applications. Relevant research studies are examined to identify the conditions under which neural networks or statistical learning methods provide advantages. The review also analyzes limitations in existing comparisons, particularly the effect of dataset characteristics, model complexity, interpretability requirements, and computational resources on model selection. Based on the identified research gap, a comparative evaluation framework is proposed in which both approaches are assessed under controlled datasets and consistent evaluation measures. The study aims to provide a clearer basis for selecting between neural-network and statistical-learning approaches according to the characteristics and requirements of a given application.

Keywords

Neural Networks; Statistical Learning; Machine Learning; Prediction; Classification; Interpretability

Introduction

Artificial intelligence and machine learning have created a wide range of methods for prediction and classification. Among these, statistical learning methods and neural networks represent two important approaches. Statistical learning provides a mathematical framework for learning relationships between input variables and outcomes, while neural networks use interconnected computational units and learned parameters to model potentially complex relationships in data. Research has examined neural networks as alternatives to traditional statistical techniques for both prediction and classification, with performance depending on factors such as sample size, model assumptions, validation methods, and evaluation measures.

Statistical learning methods are particularly useful when the structure of the problem can be expressed through relatively well-defined statistical assumptions and when understanding the relationship between variables is important. Interpretability has become an increasingly important consideration in machine learning because users may need to understand and validate how a model reaches its predictions. Recent research emphasizes the role of interpretable methods in building trust and validating data-driven findings.

Neural networks, especially deep neural networks, have demonstrated strong capabilities for learning complex patterns from data. Their statistical foundations include approximation, optimization, generalization, and training dynamics, but their highly nonlinear and often non-convex structure can make their behavior more difficult to analyze and explain. The black-box nature of deep neural networks has consequently become an important research issue, particularly in applications where decisions require transparency and human understanding.

The comparison between neural networks and statistical learning therefore cannot be based only on prediction accuracy. Factors such as data requirements, generalization, interpretability, computational cost, scalability, and the characteristics of the application can influence which approach is more appropriate. An earlier review of neural networks and statistical techniques specifically found that comparisons can be affected by sample size, whether statistical assumptions are satisfied, validation procedures, evaluation measures, and the statistical significance of performance differences.

Motivation

The motivation for this review is to examine these approaches using multiple criteria rather than treating one technique as universally superior. Understanding their respective strengths and limitations can help researchers and practitioners select models according to the requirements of a particular prediction or classification problem.

Scope of the Review

This paper focuses on comparing neural networks and statistical learning approaches with respect to:

- Prediction and classification performance
- Data requirements
- Generalization
- Interpretability
- Computational cost
- Scalability
- Suitability for different applications

The review also examines limitations in existing comparisons and identifies a research direction for evaluating both approaches under more consistent experimental conditions.

Sources used for this section

- Paliwal & Kumar, Neural networks and statistical techniques: A review of applications — directly compares neural networks with traditional statistical methods for prediction and classification.
- Nalisnick, Smyth & Tran, A Brief Tour of Deep Learning from a Statistical Perspective — discusses the statistical foundations of deep learning.
- Suh & Cheng, A Survey on Statistical Theory of Deep Learning — reviews approximation, training dynamics and generalization-related theory.
- Allen, Gan & Zheng, Interpretable Machine Learning for Discovery — discusses interpretability, validation and trustworthy machine learning.

Research Problem and Objectives

Research Problem

Neural networks and statistical learning methods are both widely used for prediction and classification, but their performance is not consistently superior across all datasets and applications. Earlier comparative research reviewed studies involving neural networks, regression, logistic regression, and discriminant analysis and found that performance comparisons can depend on factors such as sample size, model assumptions, validation methods, evaluation measures, and application characteristics.

Recent systematic evidence also suggests that the choice between neural networks and statistical models should not be based only on predictive accuracy. A 2026 systematic review comparing generalized additive models (GAMs) and neural networks across 143 papers and 430 datasets found no consistent overall superiority for either approach on common tabular-data metrics. Neural networks tended to have advantages on larger datasets and datasets with more predictors, while GAMs remained competitive on smaller datasets and offered stronger interpretability.

Therefore, the research problem addressed in this review is:

> How do neural networks and statistical learning methods differ in prediction and classification performance, data requirements, generalization, interpretability, computational cost, scalability, and suitability for different application conditions?

Research Objectives

The objectives of this research are:

- To review existing research comparing neural networks and statistical learning methods for prediction and classification.

- To compare their performance using criteria such as accuracy, generalization, computational cost, and scalability.
- To examine differences in data requirements, particularly the influence of dataset size and number of predictors.
- To analyze the interpretability of neural networks and statistical learning approaches.
- To identify the conditions under which one approach may be more suitable than the other.
- To critically evaluate limitations and inconsistencies in existing comparative studies.
- To identify a research gap concerning controlled and multi-criteria comparison of the two approaches.
- To propose a research framework for evaluating neural networks and statistical learning methods under consistent experimental conditions.

Research Questions

The review is guided by the following questions:

RQ1: How does the predictive and classification performance of neural networks compare with statistical learning methods?

RQ2: How do dataset size and number of predictors affect the relative performance of the two approaches?

RQ3: Which approach provides better interpretability?

RQ4: How do computational cost and scalability differ between the approaches?

RQ5: Under what application conditions is each approach more appropriate?

RQ6: What limitations exist in current comparative research, and what research direction n can address them?

Literature Review

The literature comparing neural networks with statistical learning methods shows that neither approach is universally superior. Performance depends on the type of data, sample size, number of predictors, model assumptions, validation procedure, and evaluation metric.

Neural Networks and Traditional Statistical Methods

Paliwal and Kumar conducted a review of 73 comparative papers examining multilayer feedforward neural networks against statistical techniques such as regression, logistic regression, and discriminant analysis. Their review focused on prediction and classification across different application areas. The authors emphasized that comparisons can be strongly influenced by sample size, assumptions of the statistical method, validation procedures, and the performance measure used. Their findings support the view that neural networks can be effective for nonlinear prediction and classification, but their superiority cannot be assumed across all applications.

Statistical Foundations of Neural Networks

Nalisnick, Smyth, and Tran examined deep learning from a statistical perspective and connected neural-network models with concepts from probability and statistics. Their review covers feedforward networks, sequential networks, and neural latent-variable models, showing that neural networks are not separate from statistical learning but have strong statistical foundations. This provides an important basis for comparing the two fields rather than treating them as completely unrelated approaches.

Suh and Cheng further reviewed the statistical theory of neural networks through three perspectives: approximation, training dynamics, and generative models. Their work discusses how neural networks can achieve useful generalization while being trained through highly non-convex optimization problems. This indicates that neural-network performance involves both statistical properties and optimization behavior.

Interpretability

Interpretability is a major difference between many statistical models and neural networks. Allen, Gan, and Zheng describe interpretable machine learning as an important area for producing human-understandable insights and validating data-driven discoveries. They emphasize that interpretability is connected not only to explanation but also to validation, reproducibility, and trust.

Recent research specifically examining deep-learning interpretability also identifies the black-box nature of deep learning as a continuing challenge. Xu and Yang's 2025 literature survey reviews different approaches for explaining deep-learning decisions and identifies interpretability evaluation and trust as active research areas.

Neural Networks vs Generalized Additive Models

A recent systematic review by Doohan, Kook, and Burke provides particularly relevant evidence for the comparison. The study examined 143 papers involving 430 datasets comparing generalized additive models (GAMs) and neural networks on real-world tabular data. The authors found no consistent overall superiority for either approach on commonly reported metrics such as RMSE, R2, and AUC. Neural networks tended to perform better on larger datasets and datasets with more predictors, while GAMs remained competitive on smaller datasets and retained their interpretability advantage.

The review also identified an important methodological issue: reporting of dataset characteristics and neural-network complexity was incomplete in a substantial portion of the literature. This limits transparency and reproducibility when comparing the two approaches.

Literature Synthesis

The reviewed studies can be synthesized into the following comparison:

Factor	**Neural Networks**	**Statistical Learning**
Nonlinear relationships	Strong capability	Depends on model
Large datasets	Often advantageous	Can remain competitive
Small datasets	May be less advantageous	Often competitive
Interpretability	Generally more difficult	Often stronger
Complex patterns	Strong	Depends on model flexibility
Computational requirements	Often higher for complex networks	Often lower for simpler models
Generalization	Depends on architecture, training and data	Depends on model assumptions and data

Model assumptions	Generally fewer explicit distributional assumptions	Often more dependent on statistical assumptions
Scalability	Strong for large/high-dimensional problems	Varies by method
Transparency	More challenging for complex networks	Often easier to explain

The evidence therefore suggests that the comparison should be context-dependent rather than accuracy-only. The recent GAM–neural-network systematic review is especially relevant because it found similar performance across many real-world tabular datasets while identifying dataset size and predictor count as factors associated with differences in performance.

Key findings from the literature

1. Neural networks are capable of modelling complex nonlinear relationships.
2. Statistical learning methods remain competitive, particularly when data are limited or interpretability is important.
3. Dataset characteristics can influence which approach performs better.

4. Neural networks have stronger computational and interpretability challenges as model complexity increases.
5. Existing research does not support the claim that one approach is universally superior.

Comparative Analysis

The literature indicates that neural networks and statistical learning methods should be compared across several dimensions rather than by accuracy alone. Recent evidence from a systematic review of 143 papers and 430 datasets found no consistent overall superiority between neural networks and generalized additive models (GAMs) on common tabular-data metrics. The relative performance was influenced by dataset size, number of predictors, application area, and neural-network complexity.

Prediction and Classification Performance

Neural networks are effective at modelling complex and nonlinear relationships. Their flexibility can be particularly useful when the relationship between predictors and outcomes is difficult to specify using a simpler statistical model.

However, statistical learning methods can remain highly competitive. The recent GAM–neural-network systematic review found no consistent evidence that either approach was universally superior across RMSE, R2, and AUC. Neural networks showed a tendency to perform better on larger datasets and datasets containing more predictors, while GAMs remained competitive on smaller datasets.

Therefore, predictive performance depends strongly on the characteristics of the dataset rather than simply on the choice between "neural network" and "statistical model."

Data Requirements

Neural networks generally benefit from larger amounts of training data, particularly when the network contains many parameters. Deep-learning literature also identifies extensive datasets and computational resources as practical challenges.

Statistical learning methods can be more practical when the available dataset is relatively small, particularly when the model incorporates appropriate assumptions or regularization.

The recent systematic comparison supports this distinction: neural networks tended to show stronger performance on larger datasets, whereas GAMs remained competitive in smaller-data settings.

Interpretability

Interpretability is one of the clearest differences between the two approaches.

Many statistical learning models provide parameters or functional relationships that can be examined directly. This can make it easier to understand how predictors contribute to an outcome.

Neural networks, particularly deep networks, can contain many layers and parameters, making their decision processes harder to interpret. Research on interpretable machine learning emphasizes that understanding model decisions is important for validation, trust, and reproducibility.

Thus:

Statistical Learning→Generally higher interpretability

Neural Networks→Generally greater explanation difficulty

This does not mean neural networks cannot be explained; rather, additional explainability techniques may be required.

Computational Cost

Neural-network training can require substantial computational resources, especially as the number of layers, parameters, and training iterations increases. Deep-learning literature identifies computational requirements as one of the practical challenges associated with neural networks.

Simpler statistical learning models can often be trained with fewer computational resources. However, computational cost depends on the particular statistical method, dataset size, and implementation, so it would be inaccurate to claim that every statistical model is computationally cheaper.

Generalization

Generalization refers to how well a trained model performs on previously unseen data.

Neural-network generalization depends on factors including:

- Dataset size
- Architecture
- Regularization
- Training procedure
- Model complexity

Statistical learning models also depend on model assumptions, regularization, feature selection, and the relationship between the training and test data.

Therefore, neither approach automatically guarantees better generalization. Evaluation should use appropriate validation or test data rather than relying only on training performance.

Scalability

Neural networks can scale effectively to large and high-dimensional datasets, which is one reason they are widely used for complex machine-learning tasks. Deep-learning reviews identify scalability as a major strength while also pointing to computational-resource requirements.

Statistical learning methods can also scale well, but scalability varies considerably between methods. A simple regression model and a computationally intensive statistical model should not be treated as equivalent.

Overall Comparison

Criterion	**Neural Networks**	**Statistical Learning**
Prediction	Strong for complex nonlinear patterns	Strong for structured relationships
Classification	Strong, especially with complex data	Strong, depending on model
Large datasets	Often advantageous	Can also perform well
Small datasets	May require careful regularization	Often competitive
Interpretability	Generally lower	Generally higher
Computational cost	Often higher for complex networks	Often lower for simpler models
Generalization	Depends on architecture and training	Depends on assumptions and regularization
Scalability	Strong for large/high-	Method-dependent

	dimensional problems	
Model assumptions	Often fewer explicit distributional assumptions	Can rely more strongly on statistical assumptions
Transparency	More difficult for complex networks	Often easier to explain

Key Comparative Finding

The strongest conclusion from the literature is not that neural networks are better than statistical learning, or vice versa. Instead, the appropriate method depends on the characteristics and requirements of the problem. The recent systematic review specifically concludes that neural networks and GAMs can be viewed as complementary approaches, with relatively modest performance differences on many tabular datasets and an interpretability advantage for GAMs.

Critical Analysis

The literature shows that comparing neural networks with statistical learning methods is more complicated than simply identifying which model produces the highest accuracy. The performance of each approach depends on dataset size, number of predictors, model complexity, validation strategy, computational resources, and the importance of interpretability. An earlier review similarly emphasized that comparative results can be affected by sample size, statistical assumptions, validation methods, and evaluation measures.

Strengths of Neural Networks

Neural networks have a major advantage in their ability to represent complex and nonlinear relationships. Their flexible architectures can learn patterns without requiring researchers to specify the complete functional relationship beforehand. Research on deep learning also connects neural networks to statistical foundations involving probability, optimization, and pattern recognition.

Their flexibility can become particularly valuable when datasets are large or contain many predictors. The recent systematic review of GAMs and neural networks found that neural networks tended to perform better on larger datasets and datasets with more predictors, although the advantage was not universal.

Weaknesses of Neural Networks

The flexibility of neural networks also creates challenges. Complex networks can require substantial computational resources and careful training. Their internal representations can be difficult to interpret, making it harder to explain why a particular prediction was produced.

Generalization is another important issue. A neural network may achieve excellent performance on training data but perform less effectively on unseen data if the training process, architecture, or dataset is inappropriate. Research on neural-network generalization shows that this problem extends beyond simple train-test performance and includes generalization across distributions, domains, tasks, and modalities.

Strengths of Statistical Learning

Statistical learning methods can provide a strong balance between predictive performance and interpretability. Models such as generalized additive models can represent nonlinear relationships while retaining a structure that is easier to examine and explain.

The recent systematic review found that GAMs remained competitive with neural networks, particularly on smaller datasets, while maintaining an interpretability advantage.

This makes statistical learning particularly attractive in applications where understanding the contribution of individual predictors is as important as obtaining accurate predictions.

Weaknesses of Statistical Learning

Statistical learning methods are not automatically superior simply because they are interpretable. Their performance can depend on the assumptions and structure of the selected model. An inappropriate model form may fail to capture complex relationships that a neural network can learn.

Therefore, the choice of a statistical model also requires careful consideration of the data-generating process, predictor relationships, validation procedure, and model assumptions. This issue was specifically emphasized in the earlier comparative review of neural networks and statistical methods.

Critical Comparison

The strongest evidence against declaring a universal winner comes from the 2026 systematic review of 143 papers and 430 datasets. It found no consistent superiority of GAMs or neural networks across commonly reported metrics such as RMSE, R2, and AUC. It also identified incomplete reporting of dataset characteristics and neural-network complexity as a limitation affecting transparency and reproducibility.

This finding has an important implication: model performance should be interpreted in context.

For example:

- A large, high-dimensional dataset may favour a neural network.
- A smaller tabular dataset may make statistical learning more competitive.

- An application requiring transparent predictor effects may favour an interpretable statistical model.
- An application involving highly complex nonlinear relationships may benefit from neural-network flexibility.
- A computationally constrained environment may favour a simpler model.

These are not absolute rules; they are conditions that should be tested empirically.

Key Critical Finding

A major weakness in the existing literature is inconsistent reporting. The recent systematic review found that important information about dataset characteristics and neural-network complexity was missing from many studies, making it harder to reproduce and fairly compare results.

Therefore, a meaningful comparison should control or clearly report:

Dataset Size+Predictor Count+Model Complexity+Validation+ Evaluation Metrics+Computational Cost

Rather than asking: “Which model is better?”

the more useful research question is:

"Under what data and application conditions does each model provide the best balance of predictive performance, generalization, interpretability, and computational efficiency?"

Research Gap

The literature shows substantial progress in comparing neural networks with statistical learning methods, but several gaps remain.

Lack of a Consistent Multi-Criteria Comparison

Many comparisons emphasize predictive performance, while other practical factors such as interpretability, computational cost, scalability, and data requirements receive less consistent treatment. A recent systematic review of GAMs and neural networks found no consistent overall superiority across RMSE, R2, and AUC, suggesting that accuracy alone is insufficient for selecting a model.

Dataset Characteristics Are Not Always Controlled

The relative performance of the approaches changes with sample size and number of predictors. The systematic review found that neural networks tended to perform better on larger datasets and datasets with more predictors, while GAMs remained competitive on smaller datasets. However, reporting of dataset characteristics and neural-network complexity was incomplete in much of the existing literature, limiting reproducibility.

Therefore, comparisons should explicitly control:

Sample Size + Predictor Count + Model Complexity

Accuracy–Interpretability Trade-off

Neural networks can provide flexible nonlinear modelling, but their complex internal structure can make interpretation difficult. Statistical and interpretable models can provide clearer relationships between predictors and outcomes. Research on interpretable machine learning identifies validation, reproducibility, and human-understandable explanations as continuing challenges.

This creates an important research question:

Can predictive performance be improved without sacrificing interpr etability?

Limited Standardization of Evaluation

Different studies use different:

- datasets,
- preprocessing methods,
- validation strategies,
- performance metrics,
- model architectures, and
- hyperparameter settings.

This makes direct comparison difficult. The recent systematic review specifically identified incomplete reporting as a limitation to transparency and reproducibility.

Identified Research Gap

Based on the literature, the main gap can be stated as:

There is a need for a controlled, reproducible, multi-criteria framework that compares neural networks and statistical learning methods under consistent datasets, preprocessing procedures, validation strategies, and evaluation metrics while simultaneously considering predictive performance, generalization, interpretability, computational cost, and scalability.

This gap provides the basis for the proposed research direction.

Proposed Research Direction

The proposed research will develop a controlled comparative evaluation framework for neural networks and statistical learning methods.

Proposed Framework

Two model groups will be evaluated:

- **Neural Network**

$$\text{Input Data} \rightarrow \text{MLP} \rightarrow \text{Prediction}$$

- **Statistical Learning**

Input Data→Statistical Model→Prediction

Both approaches will use the same datasets, preprocessing pipeline, training/test strategy, and evaluation metrics.

Evaluation Criteria

Criterion	Measurement
Prediction	RMSE, MAE, R2
Classification	Accuracy, Precision, Recall, F1, AUC
Generalization	Test/validation performance
Interpretability	Explanation clarity and model transparency
Computational Cost	Training time and resource usage
Scalability	Performance as dataset size increases
Data Requirement	Performance across different sample sizes

Experimental Design

The study can evaluate models across different dataset conditions:

Small Dataset→Medium Dataset→Large Dataset

and:

Low Dimensionality→High Dimensionality

The results can then be analyzed to determine when each approach provides the best trade-off.

Expected Contribution

Rather than producing a simple ranking such as "neural networks are better" or "statistical learning is better," the proposed study aims to produce a decision framework:

Dataset Characteristics+Application Requirements→Suitable Model

This direction is consistent with recent evidence that neural networks and GAMs can be complementary rather than universally competing approaches.

Why this research direction is relevant

Recent work also identifies continuing trade-offs between accuracy and interpretability, and newer research is exploring ways to combine neural-network predictive capability with more interpretable representations.

So the proposed research does not simply compare the two approaches it investigates the conditions under which each approach is preferable and whether their strengths can be combined.

Discussion

The comparison presented in this review shows that neural networks and statistical learning should be viewed as complementary approaches rather than as universally competing methods. Evidence from a systematic review of 143 papers and 430 datasets found no consistent superiority of neural networks or generalized additive models across RMSE, R2, and AUC. Neural networks showed an advantage more often on larger datasets and datasets with more predictors, while GAMs remained competitive on smaller datasets and provided stronger interpretability.

A major implication is that dataset characteristics should influence model selection. A model that performs well on a large, high-dimensional dataset may not provide the same advantage on a smaller tabular dataset. Therefore, model selection based only on a single accuracy value can produce misleading conclusions.

Interpretability creates another important distinction. Complex neural networks can provide strong predictive capabilities, but their internal representations can be difficult to understand. Recent surveys continue to identify the black-box nature of deep neural networks as a major challenge and emphasize the need for reliable explanation methods. Statistical learning approaches, particularly interpretable models such as GAMs, can provide a clearer relationship between predictors and predictions.

The findings also suggest that computational resources should be considered alongside predictive performance. Neural networks can become computationally demanding as model size and training requirements increase. Consequently, a small improvement in predictive performance may not justify substantially greater computational cost when the application has limited hardware or strict response-time requirements.

Another important issue is generalization. High performance on a particular dataset does not automatically mean that a model will perform well on new datasets or different populations. The comparison should therefore use appropriate validation procedures and report dataset characteristics clearly. The recent systematic review found incomplete reporting of dataset characteristics and neural-network complexity in much of the literature, which limits transparency and reproducibility.

Practical Implications

The findings suggest the following decision pattern:

Situation	**Potentially suitable approach**
Large dataset with many predictors	Neural Network
Complex nonlinear relationships	Neural Network
Small or moderate tabular dataset	Statistical Learning
Strong interpretability requirement	Statistical Learning
Limited computational resources	Simpler Statistical Model
Highly complex prediction task	Neural Network
Need for transparent predictor	Interpretable Statistical

effects	Model

These should be treated as guidelines rather than fixed rules, because the appropriate model ultimately depends on empirical evaluation.

Overall Discussion

The central finding of this research is that there is no meaningful basis for declaring one approach universally better. Instead, the choice should consider:

Performance+Generalization+Interpretability+Computational Cost+Data Requirements

A fair comparison should therefore evaluate both approaches under the same datasets, preprocessing procedures, validation strategy, and performance measures. This directly supports the proposed research direction of developing a controlled multi-criteria evaluation framework.

The discussion also reinforces the research gap identified earlier: future studies should report dataset characteristics and model configurations more consistently and evaluate performance together with interpretability and computational efficiency. This would make comparisons more reproducible and more useful for real-world model selection

Conclusion

This review compared neural networks and statistical learning methods across prediction and classification performance, data requirements, generalization, interpretability, computational cost, and scalability. The evidence indicates that neither approach is universally superior. A recent systematic review covering 143 papers and 430 datasets found no consistent overall advantage for neural networks or generalized additive models on commonly reported metrics such as RMSE, R2, and AUC. Neural networks showed greater advantages more often on larger datasets and datasets with more predictors, while GAMs remained competitive on smaller datasets and provided stronger interpretability.

The analysis also shows that model selection should depend on the characteristics of the problem. Neural networks are valuable when flexible nonlinear modelling and scalability are important, but they can require greater computational resources and may be more difficult to interpret. Statistical learning methods can provide competitive predictive performance with clearer model structures, particularly when interpretability and smaller datasets are important. Research on interpretable machine learning further emphasizes that transparency and validation are important when models are used to support scientific or practical decisions.

Generalization is another important consideration. Strong performance on training data does not necessarily guarantee reliable performance on unseen or differently distributed data. Recent research identifies sample, distribution, domain, task, and modality generalization as distinct challenges for neural networks. Therefore, comparisons should use appropriate validation procedures and should report dataset characteristics, model complexity, and evaluation conditions clearly.

The research gap identified in this review is the lack of sufficiently standardized, multi-criteria comparisons that evaluate predictive performance together with interpretability, computational cost, scalability, and data requirements. Existing evidence also shows incomplete reporting of dataset characteristics and neural-network complexity in parts of the literature, which can limit reproducibility.

Based on this gap, the proposed research direction is to evaluate both approaches under controlled and consistent experimental conditions, using the same datasets, preprocessing procedures, validation strategies, and evaluation metrics. Rather than producing a simple ranking, the study should identify which approach is most suitable under particular data and application conditions.

Therefore, the central conclusion of this review is:

There is no universal winner between Neural Networks and Statistical Learning.

The appropriate choice depends on the dataset, problem complexity, available computational resources, required interpretability, and desired predictive performance. Treating the two approaches as complementary alternatives rather than universally competing methods provides a more balanced and practically useful basis for model selection.

References

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- Generalization in neural networks: A broad survey. (2024). Neurocomputing.
- Challenges of statistical machine learning when encountered deep neural network. (2024). Handbook of Statistics, 51, 275–295.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identificationand Understanding	10%	9–10% – Clearly identifies the researchproblem, objectives,scope, and relevance to AI learning approaches.	7–8% – Research problem and objectivesare clearly identifiedwith minor omissions.	5–6% – Basic understanding oftheresearch problem with some gaps.	0–4%–Problem is unclear, inappropriate,or poorly understood.
2.Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8%–Uses relevant and credible sourceswith minor limitations.	5–6% – Uses a limitednumber of relevant sources.	0–4%–Sources are insufficient, unreliable, or largely irrelevant.
3.Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiplestudies rather than merely summarizing them.	10–12% – Good review with meaningful comparisonof major studies.	7–9%–Basic review with limited synthesisand comparison.	0–6% – Superficial reviewwith little understandingof existingresearch.

4.AnalysisofAI Techniques / Methods	15%	13–15% – Provides technically accurateand in-depthanalysis of relevant methodssuchas decision trees, statistical learning, neural networks,kernel methods, or reinforcement learning.	10–12% – Providesgood technical analysiswith minor gaps.	7–9% – Providesbasic technical description with limited analysis.	0–6% – Methods are poorly understoodor inaccurately described.
5.Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approachesbased on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison usingseveral parameters.	5–6%–Basic comparison with limited criteria.	0–4%–Littleor no meaningful comparison.
6. Critical Evaluationof Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods,and findings of existing studies.	7–8% – Provides good critical evaluationwith minor gaps.	5–6%–Some critical observations are provided butanalysisis limited.	0–4% – Mostly descriptivewith little critical evaluation.

7. Identification ofResearchGaps	10%	9–10%–Clearly identifies significant research gaps, limitations, unresolved issues, and opportunitiesfor further investigation.	7–8% – Identifies relevant research gaps withreasonable justification.	5–6% – Identifiesbasic gaps but provides limited justification.	0–4% – Researchgaps aremissingor incorrectly identified.
8. Research Insights and Recommendations	5%	5%–Provides insightful conclusionsand technically feasible recommendations for future research or applications.	4%–Provides relevant conclusionsand recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9.TechnicalReport and Referencing	10%	9–10%–Reportis well structured, technically accurate, professionally written,andusesa consistent citation/reference style.	7–8%– Well-structured reportwithminor writing or referencing issues.	5–6%– Adequatereport but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incompletereport.
10. Originality, AcademicIntegrity and Use of Sources	5%	5%– Demonstrates original synthesis, propercitation,and strong academic integrity with no inappropriate copying.	4% – Good originalityand propercitation with minor issues.	2–3% – Some originalanalysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.